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HSACT: A hierarchical semantic-aware CNN-Transformer for remote sensing image spectral super-resolution

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ABSTRACT

Hyperspectral remote sensing technology has demonstrated its spectral diagnosis advantages in numerous remote sensing observation fields. However, hyperspectral imaging is expensive and less portable compared to RGB imaging. To recover the corresponding hyperspectral image (HSI) from a remote sensing RGB image, this paper proposes a new hierarchical semantic-aware convolutional neural network (CNN)-Transformer (HSACT) for remote sensing image spectral super-resolution (SSR). Particularly, this work aims to reconstruct HSIs from RGB images within the same field of view using a lightweight semantic embedding architecture. Our HSACT consists of the following steps. First, an initial spectrum estimation module (from the RGB image to the HSI) is designed to progressively consider spectral estimation between RGB wavelength-inner and wavelength-outer information. Then, an attention-driven semantic-aware CNN-Transformer is developed to reconstruct the spatial and spectral details of HSI. Specifically, a trainable polymorphic superpixel convolution (PSConv) is proposed to capture features efficiently in the above module. Next, we introduce an information-lossless hierarchical network architecture to link the above modules and achieve end-to-end RGB image SSR through weight sharing. Experimental results on several datasets demonstrated that our HSACT outperforms traditional and advanced SSR methods. The codes of this paper are available from https://github.com/chengle-zhou/HSACT.

1. Introduction

Hyperspectral images (HSIs) are known for their fine spectral resolution [1]. In contrast to panchromatic, red-green-blue (RGB), and multispectral images (MSI), HSIs are capable of acquiring both 2-D geometric information and continuous and narrow 1-D spectral information of the imaged objects, which has attracted wide attention in various research topics (e.g., classification and change detection [2,3]) and several application fields such as food safety [4], medical diagnosis [5], and remote sensing [6–8]. Hyperspectral imaging currently faces three key challenges [9]:

- *High cost*. Imaging spectrometers (used to obtain HSIs) tend to be more expensive compared to traditional spectrometers (RGB images and MSIs) [10].
- *Dependence on weather conditions*. Imaging spectrometers are more sensitive to light and weather conditions [11].
- *Mutual constraints on spatial and spectral resolution*. Due to the design characteristics of imaging spectrometers, it is difficult for a HSI to have both high spectral and spatial resolution [12,13]. This hinders the development of applications requiring high resolution in both the spatial and spectral domains.

Over the past five years, the attention of scholars has begun to shift from the processing and quality improvement (e.g., classification [14] and denoising [15]) of HSIs to the topic of generating high spectral resolution HSIs from low spectral resolution remote sensing images (i.e., RGB images and MSIs). In essence, spectral reconstruction can break through the performance bottleneck of imaging spectrometers and further realize the potential of HSIs in various practical fields [16]. Based on existing knowledge, spectral reconstruction mainly comprises two strategies: image fusion (IF) and spectral super-resolution (SSR).

IF methods aim to achieve high spectral resolution HSIs by combining spatial details in MSIs with finer spectra in HSIs [17,18]. For example, Dian et al. proposed a spectral-spatial sparse representation approach for the fusion of an MSI and an HSI of the same area [19]. Zhou et al. designed a MSI-HSI fusion algorithm by combining linear

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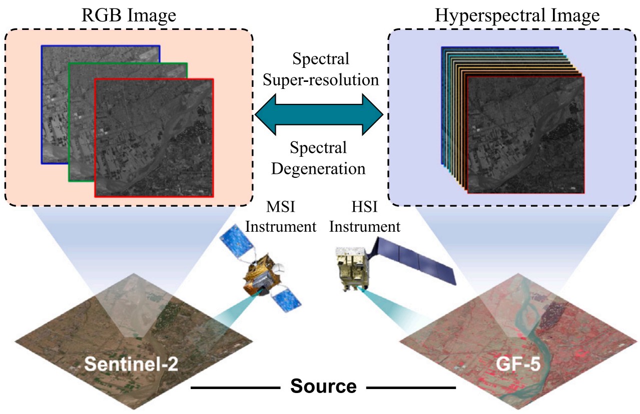

Fig. 1. Toy example illustrating the SSR process. The MSI image is collected using Sentinel-2 and the HSI image is collected using GF-5.

spectral unmixing with the local low-rank property to create spectral images with high spectral and spatial resolution [20]. Ren et al. presented a locally optimized image segmentation fusion framework for HSI super-resolution reconstruction by fusing MSIs and HSIs [21]. Zhu et al. proposed an adaptive multi-perceptual field implicit guided sampling generative adversarial network to achieve fusion of MSI-HSI image pairs [22]. In addition, an interpretable spatial–spectral reconstruction network was proposed to achieve cross-modal information fusion of spatial and spectral modes in MSI-HSI fusion [23]. Wang et al. proposed a variational probabilistic autoencoder framework, implemented by convolutional neural networks (CNNs), to fuse the spatial and spectral information contained in the MSI-HSI pair [24]. Dong et al. designed a spatial–spectral dual-optimization model-driven deep network for MSI-HSI fusion, which embeds the intrinsic generation mechanism of MSI-HSI fusion to the network [25]. Although the above fusion methods are effective in obtaining high-spatial-resolution HSIs, the model performance relies on the availability of MSI-HSI image pairs. Actually, the MSI-HSI image pairs are hardly available in practice, mainly due to two important challenges [26]:

- *Expensive acquisition of MSI-HSI pairs*. The reason is that most HSIs are used for commercial purposes and cannot be obtained for free.
- *Complex pre-processing of HSIs*. The pre-processing of HSIs includes radiometric correction, atmospheric correction, geometric correction, and denoising, among which HSI denoising is very complex on the account of the multiple types of noise.

Unlike the IF approach, SSR methods focus on producing a high-spectral-resolution HSI from the MSI (or RGB image) with super-resolution in the spectral domain [27,28] (see Fig. 1). Overall, existing SSR methods can be categorized into dictionary learning, deep network, and deep prior methods.

- Regarding dictionary learning methods, Parmar et al. [29] initially designed a sparse recovery-based reconstruction method from an RGB image to the corresponding HSI. On this foundation, a SSR method of the dictionary representation (called Arad) for each RGB pixel was proposed according to the orthogonal match pursuit algorithm in [30]. Wu et al. combined the Arad method and the A+ method proposed by Timofte et al. [31,32], introducing information about neighboring pixels. Dictionary learning methods demonstrated the possibility of recovering HSIs from MSIs, but they generally exhibit poor generalization performance.
- Regarding deep network methods, Galliani et al. earlier designed an SSR method from RGB to HSI using a CNN architecture, which proved the performance superiority of deep learning methods in SSR tasks [33]. Yan et al. presented a multi-scale deep CNN to explicitly map the input RGB image into the corresponding HSI [34]. Liu et al. introduced an RGB-to-HSI generation method based on generative adversarial networks [35]. Although deep learning networks show promising SSR results, model interpretability is a concern.
- Regarding deep prior methods, Hang et al. designed a SSR network for RGB images by taking advantage of intrinsic properties (i.e., the spectral correlation and projection properties from HSI to RGB) [36]. Chen et al. proposed a semisupervised spectral degradation constrained network to improve the spectral resolution of MSIs [37]. Wu et al. presented a holistic prior-embedded relation network for SSR of RGB images. Deep prior-based SSR methods offer significant improvements in terms of model generalizability and interpretability, but cannot be deployed in a lightweight manner [38].

Furthermore, there are several aspects of current deep learning-based SSR methods that deserve further attention:

- *Model architecture*. Encoder–decoder (EnDeCoder) architectures and sequential convolution (SeqConv) with residual connections are common, the former makes it difficult to capture spatial details and the latter exhibits inconsistent reconstruction performance across scenarios, both compared relative to hierarchical architectures. It is worth mentioning that hierarchical architectures are also concerned by scholars in the field of hyperspectral semantic segmentation.
- *Priori embedding. A priori* information is often obtained through data preprocessing and fed into the network, ignoring the deeply embedded data-driven learning paradigm in SSR networks.
- *Lightweight deployment*. Current SSR methods are more concerned with reconstruction performance than model lightweighting, making it difficult to achieve terminal deployment.
- *Real-world application exploration*. Most SSR methods perform extremely well on standard RGB-HSI datasets, but still have limitation exploration on cross-sensor data from real scenes (i.e., occurring SSR from RGB at Sentinel-2 to HSI at GF-5).

To address these limitations, we propose a new hierarchical semantic-aware CNN-Transformer (HSACT) for remote sensing image SSR. Our newly proposed method consists of three main steps. First, an initial spectral estimation module (ISE) module (from RGB to HSI) is designed to progressively consider spectral estimation between RGB wavelength-inner and wavelength-outer information. Second, an attention-driven lightweight semantic-aware CNN-Transformer is developed to reconstruct the spatial and spectral details of the HSI. We emphasize that a trainable polymorphic superpixel convolution (PSConv) is proposed to capture features efficiently in the above module. Third, we present a lightweight information-lossless hierarchical network architecture to link the above modules and achieve end-to-end RGB image SSR through weight sharing.

The key contributions of this paper can be summarized as follows:

- A novel SSR network architecture is presented to map remote sensing RGB images to the corresponding HSIs. Unlike the mainstream EnDeCoder and SeqConv architectures for SSR, it can effectively prevent information loss and enable multiplexing on the recovery side of the network in the SSR task of remote sensing images.
- Two new modules (i.e., ISE and PSConv) are introduced in the proposed HSACT method. The ISE can fully characterize the spectral-spatial content of the initial restored HSI by considering spectral estimation using both RGB wavelength-inner and wavelength-outer information. The PSConv is the first to deeply embed image structure prior into the model training paradigm instead of using it as a pre-processing part in the SSR task.

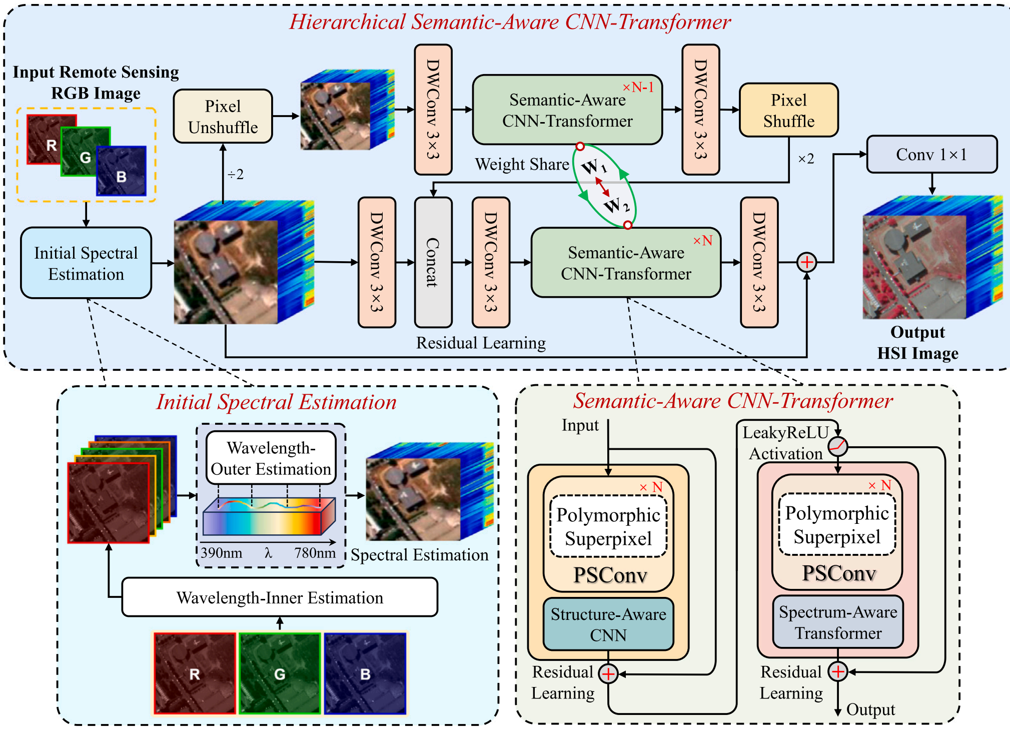

Fig. 2. Outline of the proposed HSACT method for SSR of remote sensing images.

- A new RGB-HSI dataset (termed Wenxian[1]) for the SSR task is released in this paper, in which RGB images and HSIs are from the Sentinel-2 and GF-5 satellites, respectively (see Fig. 1). Specifically, ground-truth comprises six verified land-cover classes, which demonstrated the practicality of the SSR method from a classification perspective. This dataset is the first to provide a specific practical application scenario for the SSR methods.

The remainder of this paper is organized as follows. Section 2 introduces the proposed HSACT method in detail. Section 3 presents an experimental analysis and discussions. Section 4 concludes the paper with some remarks and hints at plausible future research lines.

2. Methodology

We first provide a general overview of the proposed HSACT method and then elaborate on the model architecture and its optimization strategy. A flowchart of the proposed method is given in Fig. 2. We will describe the newly developed method in detail in five parts: (1) overview of the proposed HSACT method; (2) initial spectral estimation; (3) semantic-aware CNN-Transformer; (4) hierarchical network architecture; and (5) spectral-spatial loss function. In the following, we elaborate each of these parts.

2.1. Overview of the proposed HSACT method

Let us assume that a remote sensing RGB image is denoted as $\mathbf{R} \in \mathbb{R}^{L\times W\times c}$ and that the corresponding HSI is given by $\mathbf{H} \in \mathbb{R}^{L\times W\times C}$, where L and W respectively refer to the height and width of $\mathbf{R}$ and $\mathbf{H}$, where c and C are the number of spectral channels of both images, respectively ($c = 3 \ll C$).

From the perspective of spectral imaging, the value $I_s(x, y)$ of the RGB camera at the spatial position (x, y) can be defined as follows:

$$I_s(x, y) = \int_{\Lambda} E(x, y, \lambda)A_s(\lambda)d\lambda \tag{1}$$

where $s \in \{\mathbf{S}_r, \mathbf{S}_g, \mathbf{S}_b\}$ refers to the color channel in the RGB image, $E(x, y, \lambda)$ represents the spectral radiance in the wavelength range λ at position (x, y), $A_s(\lambda)$ is the spectral response function (SRF) at spectral channel s from the HSI to the RGB image, and Λ stands for the wavelength interval. In real scenarios, the wavelength interval is often discretely sampled, and thus Eq. (1) can be redefined as follows:

$$I_s(x, y) = \sum_{n} E(x, y, \lambda_n)A_s(\lambda_n) \tag{2}$$

where n represents the number of spectral bands in the HSI.

According to Eq. (2), the conversion relationship between $\mathbf{R}$ and $\mathbf{H}$ can be described as follows:

$$\mathbf{R} = \mathbf{A}\mathbf{H}, \quad \mathbf{A} \in \mathbb{R}^{c\times C} \tag{3}$$

where $\mathbf{A} \in \mathbb{R}^{c\times C}$ is the spectral response of the RGB camera. Existing deep learning-based SSR methods overcome the ill-posed problem of the calculation of $\mathbf{A}$ by learning a mapping function $f(\mathbf{R}; \theta)$ that reconstructs $\mathbf{H}$ from $\mathbf{R}$. In order to achieve this goal, we can minimize the following objective function:

$$\arg\min_{\theta} \mathcal{L}(f(\mathbf{R}; \theta), \mathbf{H}) \tag{4}$$

where $\mathcal{L}(\cdot)$ represents a loss function. In this work, we focus on the hierarchical network architecture to design the mapping model $f(\cdot)$ instead of the traditional EnDeCoder and SeqConv design modules, and embed the $\mathbf{R}$ structure into $f(\cdot)$ in an end-to-end learnable way. Finally, we optimize the model with the spectral-spatial loss function. The objective function of the proposed HSACT method is given by:

$$\begin{aligned}\arg\min_{\theta} \{&\mathcal{L}_{spe}(f(\mathbf{R}; \mathcal{P}_s(\mathbf{R}); \theta), \mathbf{H}) \\ &+ \alpha \cdot \mathcal{L}_{spa}(f(\mathbf{R}; \mathcal{P}_s(\mathbf{R}); \theta), \mathbf{H})\}\end{aligned} \tag{5}$$

where $\mathcal{L}_{spe}$ and $\mathcal{L}_{spa}$ respectively refer to the spectral and spatial loss function, α stands for a trade-off parameter of spectral and spatial information, $\mathcal{P}_s(\mathbf{R})$ is a structural prior embedding provided by the designed PSConv module (the details of this module are presented in Section 2.3.1).

[1] Available online at: https://github.com/chengle-zhou/RGB-HSI_Wenxian.

2.2. Initial spectral estimation

Given an RGB image $\mathbf{R} \in \mathbb{R}^{L\times W\times c}$, the initial HSI $\mathbf{H} \in \mathbb{R}^{L\times W\times C}$ corresponding to $\mathbf{R}$ are obtained by a two-stage progressive spectral estimation performed on $\mathbf{R}$. The first stage is to utilize the averaging operation between spectral channels to estimate the wavelength-inner channel information between the red and green channels (i.e., $\mathbf{S}_{rg}$) and the green and blue channels (i.e., $\mathbf{S}_{gb}$), respectively, and the raw and the estimation spectral channels can be combined as fine spectral information $\mathbf{S}_R$, which can be described as follows:

$$\begin{aligned} \mathbf{S}_{rg} &= \left(\mathbf{S}_r + \mathbf{S}_g\right)/2 \\ \mathbf{S}_{gb} &= \left(\mathbf{S}_g + \mathbf{S}_b\right)/2 \\ \mathbf{S}_R &= \{\mathbf{S}_r, \mathbf{S}_{rg}, \mathbf{S}_g, \mathbf{S}_{gb}, \mathbf{S}_b\} \end{aligned} \tag{6}$$

where $\mathbf{S}_r$, $\mathbf{S}_g$, and $\mathbf{S}_b$ respectively refer to the red, green and blue spectral channels of $\mathbf{R}$. The second stage is to design a depthwise convolution (DWConv)-based spectral learning module $\hat{\mathbf{H}}_i$ for wavelength-outer estimation. The procedure is as follows:

$$\hat{\mathbf{H}}_i = \begin{cases} \mathrm{DWConv}_R(\mathbf{S}_R), & \lambda_R \in [\lambda_s, \frac{1}{3}\lambda) \\ \mathrm{DWConv}_G(\mathbf{S}_R), & \lambda_G \in [\frac{1}{3}\lambda, \frac{2}{3}\lambda] \\ \mathrm{DWConv}_B(\mathbf{S}_R), & \lambda_B \in (\frac{2}{3}\lambda, \lambda_e] \end{cases} \tag{7}$$

where $i = \{\lambda_R, \lambda_G, \lambda_B\}$ is the ith wavelength group, λ_s and λ_e are the left and right endpoints of the wavelength interval λ, respectively, λ_R, λ_G, λ_B are the three equal molecular intervals of the wavelength interval λ in sequence, and the DWConv module contains a 1×1 Conv layer followed by mirror reflection filling, a 3×3 Group Conv layer, and a 1×1 Conv layer. The $\hat{\mathbf{H}}_i$, as the initial spectral estimation feature, can be gradually restored to a HSI by subsequent networks.

2.3. Semantic-aware CNN-Transformer

The lower rightmost part of Fig. 2 provides an outline of the newly designed semantic-aware CNN-Transformer (SACT) architecture. Specifically, this architecture consists of a structure-aware CNN (SACNN) module for reconstructing structural details and a spectrum-aware Transformer (SAFormer) module for recovering the spectral information. Additionally, each module capture features by means of the proposed PSConv module. The architectural details of these three modules are presented below.

2.3.1. PSConv

Fig. 3 shows the designed PSConv module in detail. A superpixel semantic module $\mathcal{F}_s$ is first applied based on DWConv and Conv modules. The expression of this module is given as follows:

$$\mathcal{F}_s = \mathrm{Softmax}(\mathrm{Conv}_{1\times 1}(\mathrm{DWConv}(\mathbf{X}; N_s))) \tag{8}$$

where $\mathrm{Conv}_{1\times 1}$ is a convolution layer with 1×1 kernel size and N_s is a preset number of superpixels. The learnable approach in [39] is used to extract superpixel semantic information from the image in this work (see Fig. 4). This is different from traditional pre-processing superpixel segmentation and embedding. Then, the adaptively weighted superpixel semantic-driven spectral feature $\mathcal{F}_{\text{spe}}$, the spatial feature $\mathcal{F}_{\text{spa}}$, and the spectral-spatial feature ($\mathcal{F}_{\text{spe-spa}}$) are obtained as follows:

$$\mathcal{F}_{\text{spe}} = \mathrm{SpeConv}(\mathbf{X} + w_1 \cdot \mathcal{F}_s) \tag{9}$$

$$\mathcal{F}_{\text{spa}} = \mathrm{SpaConv}(\mathbf{X} + w_2 \cdot \mathcal{F}_s) \tag{10}$$

$$\mathcal{F}_{\text{spe-spa}} = \mathrm{SpeSpaConv}(\mathbf{X} + w_3 \cdot \mathcal{F}_s) \tag{11}$$

where $\mathrm{SpeConv}(\cdot)$ is a convolution layer with 1×1 kernel size, $\mathrm{SpaConv}(\cdot)$ is a DWConv layer with kernel size 3×3, $\mathrm{SpeSpaConv}(\cdot)$ is a convolution layer with 1×1 kernel size followed by a 3×3 DWConv layer, and w_1, w_2, and w_3 are learnable adaptive weights, respectively. It is worth mentioning that SpeConv aims to learn deep semantic features for recovering spectral information, SpaConv focuses on spatial detail recovery through deep aggregation of contextual information, and SpeSpaConv comprehensively performs spectral and spatial recovery. Furthermore, this multimodal aggregation operator significantly enhances the feature extraction capability and enriches the feature learning space. After that, an adaptive weighted feature is used as the output $\mathcal{F} \in \mathbb{R}^{C\times L\times W}$ of PSConv, and the expression is as follows:

$$\mathcal{F} = w_1 \cdot \mathcal{F}_{\text{spe}} + w_2 \cdot \mathcal{F}_{\text{spa}} + w_3 \cdot \mathcal{F}_{\text{spe-spa}} \tag{12}$$

2.3.2. SACNN

In order to deeply capture and learn the spatial details that are beneficial to HSI reconstruction, the intermediate features of the PSConv module are fed into the SACNN module. To simplify the description, the input features of SACNN are denoted as $\mathbf{F}_{in}$. The expression is as follows:

$$\mathbf{W}_{\text{str}} = f_s(\mathrm{Conv}_{1\times 1}\mathcal{C}(\mathrm{MaxPool}(\mathbf{F}_{in}), \mathrm{AvgPool}(\mathbf{F}_{in}))) \tag{13}$$

where $\mathbf{W}_{\text{str}}$ denotes the structure attention, $f_s(\cdot)$ refers to the Sigmoid function, $\mathcal{C}(\cdot)$ represents the concatenation operation, and $\mathrm{MaxPool}(\cdot)$ and $\mathrm{AvgPool}(\cdot)$ are max pooling and average pooling, respectively. Here, we have:

$$\tilde{\mathbf{F}}_{\mathbf{in}} = \mathbf{F}_{in} \otimes \mathbf{W}_{\text{str}}, \ \mathbf{W}_{\text{str}} \in \mathbb{R}^{L\times W} \tag{14}$$

where $\otimes$ stands for the matrix dot product.

2.3.3. SAFormer

To further obtain the long-range dependence between spectral information, so as to improve the spectral quality of reconstructed HSI. The proposed SAFormer module is applied to features achieved by the PSConv. First, it obtains the corresponding query $\mathbf{Q} \in \mathbb{R}^{C\times LW}$, key $\mathbf{K} \in \mathbb{R}^{C\times LW}$, and value $\mathbf{V} \in \mathbb{R}^{C\times LW}$ through the expansion operation of spatial dimension of the input feature $\mathbf{F}_{in} \in \mathbb{R}^{C\times L\times W}$. The process is executed as follows:

$$\mathbf{W}_{\text{spe}} = \mathrm{Softmax}(f_{\text{SpeMax}}(\mathbf{Q}\mathbf{K}^T) - \mathbf{Q}\mathbf{K}^T) \tag{15}$$

where $\mathbf{W}_{\text{spe}}$ is a spectral attention term and $f_{\text{SpeMax}}(\cdot)$ takes the maximum value followed by a vector broadcast operation.

$$\tilde{\mathbf{F}}_{\mathbf{in}} = (\beta \mathbf{W}_{\text{spe}}) \otimes \mathbf{V}, \ \mathbf{W}_{\text{spe}} \in \mathbb{R}^{C\times C} \tag{16}$$

where β refers to an adaptive adjustment coefficient learned with the network. Subsequently, the $\tilde{\mathbf{F}}_{\mathbf{in}} \in \mathbb{R}^{C\times LW}$ is unfolded as $\tilde{\mathbf{F}}_{\mathbf{in}} \in \mathbb{R}^{C\times L\times W}$ by performing a matrix transform.

2.4. Hierarchical network architecture

We designed a hierarchical network (HiNet) architecture to connect the above-mentioned ISE and SACT modules, and thus achieving the reconstructed HSI $\hat{\mathbf{H}}$. The advantages of our layered network architecture can be summarized as follows: (1) feature decoder reuse, which promotes the lightweight nature of the network; and (2) enhanced feature capture abilities of the receptive field without losing the main spatial detail information. The procedure is defined as follows:

$$\begin{aligned} &\tilde{\mathbf{F}}_{l_i} = f_{\text{PSF}}(f_{l_i}(\mathrm{SACT}(\mathbf{F}_{l_i}; \Theta_{l_i}))), \mathbf{F}_{l_i} = f_{\text{PSUF}}(\mathbf{F}_{l_{i-1}}) \\ &\qquad \vdots \\ &\tilde{\mathbf{F}}_{l_2} = f_{\text{PSF}}(f_{l_2}(\tilde{\mathbf{F}}_{l_i}, \mathrm{SACT}(\mathbf{F}_{l_2}; \Theta_{l_2}))), \mathbf{F}_{l_2} = f_{\text{PSUF}}(\mathbf{F}_{in}) \\ &\hat{\mathbf{H}} = \mathrm{Conv}_{1\times 1}(\mathbf{F}_{in} + f_{l_1}(\tilde{\mathbf{F}}_{l_2}, \mathrm{SACT}(\mathbf{F}_{in}; \Theta_{l_1}))) \end{aligned} \tag{17}$$

where $\mathbf{F}_{l_i}$ is the input feature of the l_i-th layer, $\tilde{\mathbf{F}}_{l_i}$ is the output feature of the l_i-th layer, f_{PSF} refers to the PixelShuffle operation [40] with lossless spatial downsampling information, f_{PUSF} is the reverse operation of f_{PSF}, $\mathrm{SACT}(\cdot)$ stands for the designed SACT module, and Θ_{l_i} denotes the parameters of the SACT. It is worth noting that in order

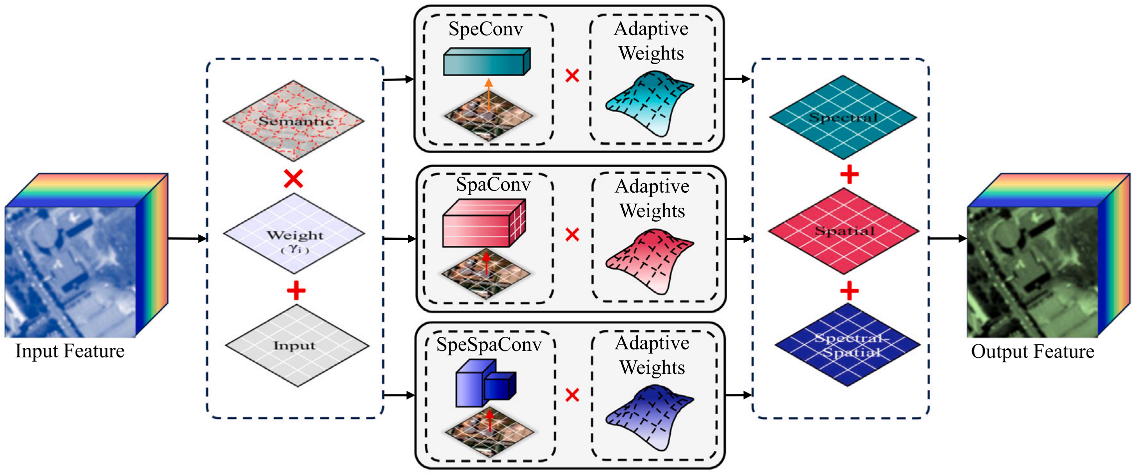

Fig. 3. Illustration of the newly designed PSConv module in the HSACT framework.

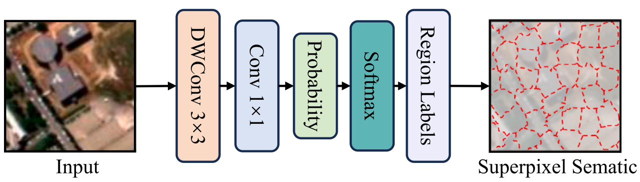

Fig. 4. Superpixel semantic module.

to improve the lightweight of the HiNet, a weight sharing strategy is introduced to the HiNet, that is, the SACT module at auxiliary layer ($l_i > 1$) directly uses the weight parameters of that at the main layer ($l_i = 1$). The formula is expressed as follows:

$$\mathrm{SACT}(\mathbf{F}_{in};\Theta_{l_1}) \overset{\text{Weight Sharing}}{\longleftrightarrow} \mathrm{SACT}(\mathbf{F}_{l_i};\Theta_{l_i}),\quad l_i \geq 1 \tag{18}$$

where $\mathbf{F}_{in}$ represents the feature map initially received by the network. We emphasize that the weight contribution strategy can significantly reduce the amount of network parameters and promote model lightweight compared to using independent parameters for each layer.

2.5. Spectral-spatial loss function

In order to make the proposed HSACT method achieve its intended purpose, we design a spectral-spatial loss function. Specifically, the spectral-spatial loss (SSLoss) function in this work is the first one developed for SSR tasks, and it has two main purposes: (1) to measure the similarity of the spectral radiance in the wavelength range of a single pixel; (2) to constrain the spatial fidelity of the spectral image at each sampling wavelength. The final loss function can be defined as:

$$\begin{aligned}\mathcal{L} &= \mathcal{L}_1 + \frac{\alpha}{2}\left(\mathcal{L}_{spe} + \mathcal{L}_{spa}\right)\\ &= \mathcal{L}_1 + \alpha\mathcal{L}_{\text{spe-spa}}\end{aligned} \tag{19}$$

where $\mathcal{L}_1$ is the L_1 loss function, α refers to a balance parameter for both terms, and $\mathcal{L}_{\text{spe-spa}}$ is the spectral-spatial loss function designed in this work. The details are given as follows:

$$\mathcal{L}_1 = \frac{1}{WLC}\sum\left|\mathbf{H} - \hat{\mathbf{H}}\right| \tag{20}$$

where W, L, and C are the width, height, and number of spectral channels of the HSI, $\mathbf{H} \in \mathbb{R}^{L\times W\times C}$ is the reference HSI, and $\hat{\mathbf{H}} \in \mathbb{R}^{L\times W\times C}$ is the reconstructed HSI. Furthermore, we need to design the spectral loss function $\mathcal{L}_{spe}$ and the spatial loss function $\mathcal{L}_{spa}$.

For the spectral loss function $\mathcal{L}_{spe}$, We measure the loss between each pixel feature (i.e., $1\times C$ spectra from $W \times L$ pixels) using spectral angle metric. Specifically, the reference HSI and the reconstructed HSI are firstly expanded along the spectral dimension, that is, $\mathbf{H}_{spe} \in \mathbb{R}^{LW\times C}$ and $\hat{\mathbf{H}}_{spe} \in \mathbb{R}^{LW\times C}$ are obtained by unfolding $\mathbf{H} \in \mathbb{R}^{L\times W\times C}$ and $\hat{\mathbf{H}} \in \mathbb{R}^{L\times W\times C}$, respectively. The spectral angle between each pixel of the $\mathbf{H}_{spe}$ and $\hat{\mathbf{H}}_{spe}$ is calculated as follows:

$$\mathcal{L}_{spe} = \frac{1}{WL}\sum\cos^{-1}\left(\frac{\left\langle \mathbf{H}_{spe}, \hat{\mathbf{H}}_{spe}\right\rangle}{\left\|\mathbf{H}_{spe}\right\|\left\|\hat{\mathbf{H}}_{spe}\right\|}\right) \tag{21}$$

where $\langle\cdot\rangle$ represents the dot product operation of the spectral radiance of each pixel, and $\|\cdot\|$ is the square root of the squared accumulation of the spectral radiance of each pixel, and $1/WL$ acts as the mean spectral angle of all pixels.

Moreover, we also consider the spatial loss between each spectral image (i.e., spectral images of size $W \times L$ from C channels) using the spectral angle metric. The spectral angle of the image for each channel is obtained as follows:

$$\mathcal{L}_{spa} = \frac{1}{C}\sum\cos^{-1}\left(\frac{\left\langle \mathbf{H}_{spa}, \hat{\mathbf{H}}_{spa}\right\rangle}{\left\|\mathbf{H}_{spa}\right\|\left\|\hat{\mathbf{H}}_{spa}\right\|}\right) \tag{22}$$

where $\mathbf{H}_{spa} \in \mathbb{R}^{C\times LW}$ and $\hat{\mathbf{H}}_{spa} \in \mathbb{R}^{C\times LW}$ is achieved by a matrix transformation of $\mathbf{H} \in \mathbb{R}^{L\times W\times C}$ and $\hat{\mathbf{H}} \in \mathbb{R}^{L\times W\times C}$, respectively. $1/C$ is used to take the mean spectral angle of the spectral images of all channels. Note that the number of pixels (i.e., LW) is considered as the feature dimension during the $\mathcal{L}_{spa}$ calculation.

3. Experiments

3.1. Dataset description

In order to verify the SSR performance of the proposed HSACT method, we perform experiments on three RGB-HSI datasets: University of Pavia, Washington DC, and our newly released Wenxian dataset. Their RGB image and hyperspectral false color composite are shown in Fig. 5.

(1) The first one was obtained over the campus of the University of Pavia, Italy, by the Reflective Optics System Imaging Spectrometer (ROSIS-3). The size of this image is 610×340 pixels, the spatial resolution of each pixel is 1.3 m, and the spectral resolution is 340–860 nm. In our experiments, 13 noisy bands are eliminated from the original set of 115 spectral bands, and we used the SRF of Sentinel-2 to generate corresponding RGB images from the remaining bands. The RGB image and false color image of the HSI are presented in Fig. 5(a). Note that

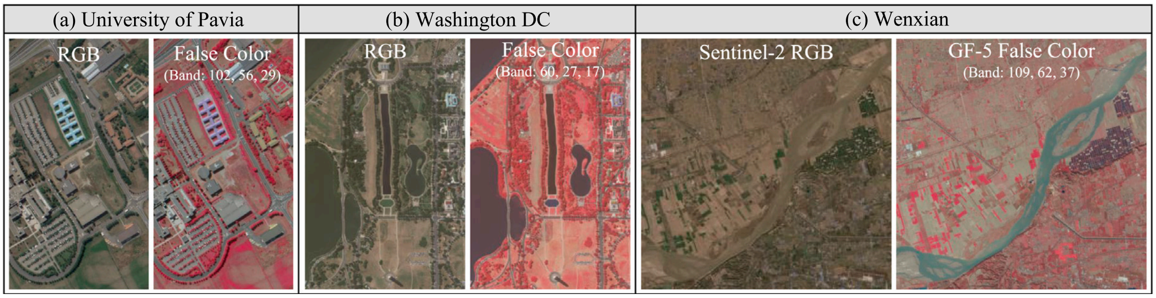

Fig. 5. RGB images and false color image associated to the considered HSIs. (a) University of Pavia scene. (b) Washington DC scene. (c) Wenxian scene.

the false color image of this scene is a composite of bands 102, 56, and 29.

(2) The second RGB-HSI dataset was collected by the Hyperspectral Digital Image Collection Experiment (HYDICE) sensor over the Washington DC Mall. 78 spectral bands from the 400–1000 nm spectral range are selected as the HSI, and the corresponding RGB images are obtained using the Sentinel-2 SRF. The dataset contains 480 scan lines and 307 pixels in each scan line. A false-color composite of the HSI and the corresponding RGB data are respectively shown in Fig. 5(b). Note that the false color image of this scene is a composite of bands 60, 27, and 17.

(3) The third dataset is our newly released Wenxin RGB-HSI dataset. The RGB image and the HSI are respectively captured by Sentinel-2 (on November 14, 2019) and GF-5 (on November 10, 2019) over Wenxian, Zhengzhou, Henan Province, China. Note that, Sentinel-2 data contains 12 spectral bands with three spatial resolutions (i.e., 10, 20, and 60 m). We selected the R, G, and B bands with 10 m spatial resolution to construct an RGB image with 30 m spatial resolution based on mean aggregation downsampling. GF-5 data are collected by the Advanced Hyperspectral Imager (AHSI) sensor, with an image spatial resolution of 30 m and 330 spectral bands from the spectral range 400–2500 nm. For the Wenxian dataset, 78 spectral bands with high signal-to-noise ratio (400–730 nm) in the V-NIR spectral range were selected as the HSI, and the ground-truth (comprising 6 land-cover classes) is made available by us. The data are available online from https://github.com/chenglezhou/RGB-HSI_Wenxian. The RGB image from Sentinel-2 and the false-color composite image from GF-5 are respectively shown in Fig. 5(c). Note that the false color image of this scene is a composite of bands 109, 62, and 37.

In addition, we randomly selected 1024 image patches of 32 × 32 size from each dataset as the training set for the training phase. In the inference phase, we used the RGB of each data set directly as input.

3.2. Performance metrics

In this work, the reference HSI $\mathbf{H}$ and the reconstructed HSI $\hat{\mathbf{H}}$ are converted into 8-bits to calculate quantitative performance. We consider the five quality metrics presented below:

3.2.1. Cross coefficient (CC)

This metric is formulated as

$$\mathrm{CC} = \frac{1}{C}\sum_{k=1}^{C} \frac{\mathrm{cov}(\mathbf{H}^k, \hat{\mathbf{H}}^k)}{\sqrt{\mathrm{var}(\mathbf{H}^k)}\sqrt{\mathrm{var}(\hat{\mathbf{H}}^k)}} \tag{23}$$

where $\mathrm{cov}(\cdot)$ is the covariance operation and $\mathrm{var}(\cdot)$ is the variance function. $\mathbf{H}^k$ and $\hat{\mathbf{H}}^k$ refer to the kth spectral band of $\mathbf{H}$ and $\hat{\mathbf{H}}$, respectively.

3.2.2. Spectral angle mapper (SAM)

This metric is expressed as

$$\mathrm{SAM} = \frac{1}{LW}\sum_{z=1}^{LW}\left(\frac{arc\cos\left\langle \mathbf{H}^z, \hat{\mathbf{H}}^z\right\rangle}{\|\mathbf{H}^z\|\left\|\hat{\mathbf{H}}^z\right\|}\right) \tag{24}$$

where $arc\cos(\cdot)$ is arc-cosine function. $\mathbf{H}^z$ and $\hat{\mathbf{H}}^z$ refer to the zth pixel of $\mathbf{H}$ and $\hat{\mathbf{H}}$, respectively.

3.2.3. Peak signal-to-noise ratio (PSNR)

This metric is defined as

$$\mathrm{PSNR} = -\frac{10}{LW}\sum_{z=1}^{LW}\log\left(\mathbf{H}^z, \hat{\mathbf{H}}^z\right)^2 \tag{25}$$

3.2.4. Structural similarity (SSIM)

This metric is formulated as

$$\mathrm{SSIM} = \frac{1}{C}\sum_{k=1}^{C}\frac{\left(2A(\mathbf{H}^k)A(\hat{\mathbf{H}}^k) + C_1\right)\left(2\mathbf{S} + C_2\right)}{\left(\mathbf{M}_1 + C_1\right)\left(\mathbf{V}_1 + C_2\right)} \tag{26}$$

where $A(\mathbf{H}^k)$ and $A(\hat{\mathbf{H}}^k)$ denote the mean of the patch in the kth band of $\mathbf{H}$ and $\hat{\mathbf{H}}$, respectively. $\mathbf{S} = \mathrm{cov}(\mathbf{H}^k, \hat{\mathbf{H}}^k)$ refers to the covariance between the patches in the kth band of both $\mathbf{H}$ and $\hat{\mathbf{H}}$, respectively, $\mathbf{M}_1 = A(\mathbf{H}^k)^2 + A(\hat{\mathbf{H}}^k)^2$, and $\mathbf{V}_1 = \mathrm{var}(\mathbf{H}^k)^{1/2} + \mathrm{var}(\hat{\mathbf{H}}^k)^{1/2}$.

3.2.5. Global relative error of synthesis (ERGAS)

This metric is given by

$$\mathrm{ERGAS} = 100 \cdot \gamma \cdot \sqrt{\frac{1}{C}\sum_{k=1}^{C}\frac{\mathrm{MSE}(\mathbf{H}^k, \hat{\mathbf{H}}^k)}{A(\mathbf{H}^k)^2}} \tag{27}$$

where $\mathrm{MSE}(\cdot)$ stands for the mean square error function and γ as a ratio parameter is set to 1.0 in this work.

The value range of SAM is [0, 1] (the larger the value, the higher the spectral distortion). CC and SSIM are utilized to investigate the spatial fidelity of the HSI, with an ideal value of 1. ERGAS and PSNR are used to verify the quality of the reconstructed HSI. The ideal value of ERGAS is 0. The larger the PSNR value, the better the SSR quality.

3.3. Competitive methods

To objectively investigate the performance of the proposed HSACT method on the SSR task for RGB remote sensing images, seven comparison methods (mainstream and state-of-the-art) are employed to test the SSR performance for all methods. The details of these methods are as follows.

- *Sparse coding method* (Arad) [30]: It first leverages the HSI before creating a sparse dictionary of hyperspectral signatures and RGB projections, and then the HSI is recovered from the corresponding RGB image according to spectral estimation with the sparse dictionary prior.

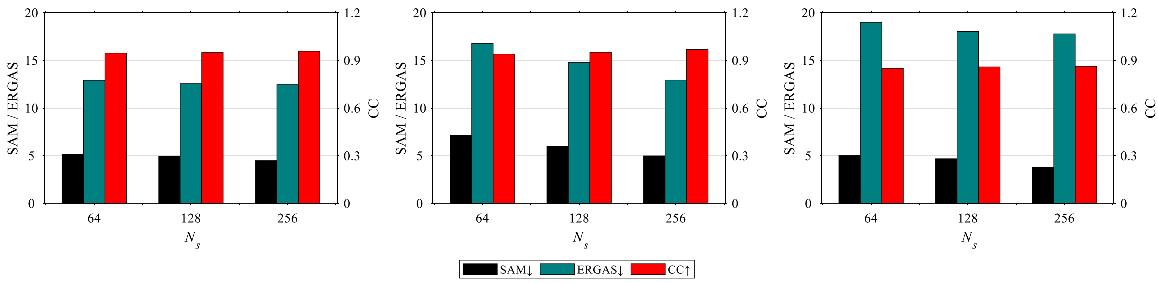

Fig. 6. Analysis of the influence of the preset number of superpixel on the SSR performance of HSACT for different datasets. (a) University of Pavia. (b) Washington DC. (c) Wenxian.

- *Joint sparse and low-rank learning* (J-SLOL) [41]: This method aims at enhancing the MSI by jointly learning low-rank MSI-HSI dictionary pairs from overlapped regions.
- *Spatial–spectral HSI prior SSR network* (HSRNet) [42]: An optimization-driven CNN for HSI reconstruction with a deep spatial–spectral prior; it can repeatedly update the intermediate HSI in an end-to-end manner.
- *Mask-guided spectral-wise transformer* (MST++) [43]: This approach belongs to a Transformer-based encoder–decoder architecture; it aims to overcome the problem that existing CNN-based methods cannot effectively capture spectral-level similarity and field-scale dependence.
- *Unmixing-guided convolutional transformer* (UGCT) [44]: A fusion network of Transformer and residual blocks guided by the unmixing paradigm; it captures local and non-local features.
- *Low-rank tensor reconstruction network* (LTRN) [45]: This algorithm treats the features of HSIs as 3-D tensors with low-rank properties, adaptive low-rank prior learning, and attempts to recover HSI from RGB images by adaptive low-rank prior learning.
- *Holistic prior-embedded relation network* (HPRN) [38]: A holistic prior-embedded relation network that integrate comprehensive priors to regularize and optimize the solution space of SSR.

3.4. Parameter configuration

The details of the setup of the comparison methods and the proposed HSACT are as follows:

For the deep learning comparison methods (StiNet, HSRNet, MST++, HDNet, and UGCT), the number of training epochs, learning rate, and optimizer in the training phase are set to 200, 1×10^{-4}, and Adam. Furthermore, other special parameters related to the default settings given by the author. Arad and J-SLOL methods use the original settings of the open source code. For a fair and objective comparison, the above methods and the proposed HSACT are all executed under the same experimental conditions. Note that this is only a general description; the details can be found in the corresponding literature.

For the proposed HSACT method, the sample batch, learning rate and number of training rounds are respectively set to 128, 1×10^{-3}, and 200 in the model training phase. The number of layers of the hierarchical structure network is preset to $l = 2$ to keep the model lightweight. The designed spectral-spatial loss function and Adam optimizer are used to obtain the optimal weight hyperparameters of the network. Furthermore, we explored the effect of hyperparameters (i.e., the number of superpixels) in the superpixel semantic module on the performance of the HSACT method with the University of Pavia, Washington DC, and Wenxian datasets. As shown in Fig. 6, the number of preset superpixels is set to 64, 128, and 256, and three metrics (i.e., CC, SAM and ERGAS) are employed to analyze the SSR performance of the HSACT. It can be clearly observed from Fig. 6 that there is a significant improvement in the performance of the HSACT as the number of superpixel blocks increases, which can be easily seen in Figs. 6(b) and 6(c). This phenomenon also illustrates two advantages of the superpixel semantic embeddings: (1) within a reasonable range, the more superpixel blocks, the smaller the local superpixel blocks, and the more correlated the internal pixels, enhancing the representation ability of the HSACT method for spectral-spatial features; and (2) the superpixel semantics can efficiently guide the deep network to perform feature learning from a local viewpoint, which can improve the SSR capability of the HSACT. Through comprehensive consideration of spatial and spectral reconstruction performance metrics (SAM, ERGAS, and CC), 256 is set as the default number of superpixel of the HSACT.

3.5. Analysis of ablation experiments

In this section, we demonstrate the effectiveness of the HSI initial estimation (i.e., the ISE module), feature learning (i.e., the PSConv, SACNN, SAFormer, and HiNet modules), and the spectral-spatial loss function (i.e., the SSLoss) of the proposed HSACT method via ablation experiments on the University of Pavia dataset. The results are presented in Table 1, where SpeConv refers to a baseline network that contains only convolutional layers with plenty of 3 × 3 DWConv and residual blocks for spatial and spectral recovery.

3.5.1. Initial spectral estimation

As shown in Table 1, it can be observed that the CC, SAM, and ERGAS of the HSACT method are 0.9590, 4.5034, and 21.473, respectively. When the ISE module is eliminated from the HSACT framework, the three metrics decrease to 0.9176, 6.5876, and 19.216, respectively. As described in Section 2.2, the ISE first enriches the original channel features by estimating the wavelength-inner features (i.e., $\mathbf{S}_{rg}$ and $\mathbf{S}_{gb}$) using the averaging operation, followed by inferencing the wavelength-out spectral features by utilizing the convolution function grouped on the five base spectral channels, which results in the fact that the spatial and spectral features approximate the HSI. The degradation of the spectral similarity, spatial fidelity, and spectral-spatial composite metrics described above reflects the effectiveness of the ISE module for the initial HSI estimation, and it demonstrates the contribution of the ISE module to the SSR task of HASCT.

3.5.2. Modules for feature learning

The feature learning modules and the HiNet (i.e., hierarchical architecture) are executed for ablation analysis and the experimental results are shown in Table 1. The PSConv plays an important role as a key plug-and-play feature recovery module. The spectral recovery performance is degraded by 32.79% (as indicated by the SAM metric) and the spatial fidelity is degraded by 3.12% (as indicated by the CC metric) when the PSConv is ablated from the HSACT, which implies that the PSConv module with superpixel semantics has a significant contribution to HSI reconstruction, especially in terms of spectral recovery. SACNN and SAFormer are attention mechanisms based on spatial and spectral features, respectively, and they are able to perceive spatial and spectral details. With the ablation of the two modules, it can be observed that the SSR performance of the HSACT undergoes a decline. This demonstrates the effectiveness of the SACNN and SAFormer modules

Table 1
Ablation study for the proposed HSACT method on the University of Pavia scene. The bold type is used for the optimal indicator, and the italic underline is used for the sub-optimal indicator.

Method	ISE	PSConv	SACNN	SAFormer	HiNet	SSLoss	Metric				
							CC↑	PSNR↑	SSIM↑	SAM↓	ERGAS↓
SeqConv	✗	✗	✗	✗	✗	✗	0.9176	31.460	0.8770	6.5876	19.216
HSACT (w/o)	✗	✓	✓	✓	✓	✓	0.9345	34.422	0.8940	5.6772	15.640
	✓	✗	✓	✓	✓	✓	0.9292	31.170	0.8714	5.9800	18.201
	✓	✓	✗	✓	✓	✓	0.9506	34.107	0.9123	5.1083	14.505
	✓	✓	✓	✗	✓	✓	0.9427	33.487	0.9029	5.2748	15.379
	✓	✓	✓	✓	✗	✓	0.9518	34.797	0.9131	5.0200	13.937
	✓	✓	✓	✓	✓	✗	*0.9556*	*34.906*	*0.9213*	*4.8040*	*13.628*
HSACT	✓	✓	✓	✓	✓	✓	**0.9590**	**36.660**	**0.9291**	**4.5034**	**12.473**

• w/o refers to without.

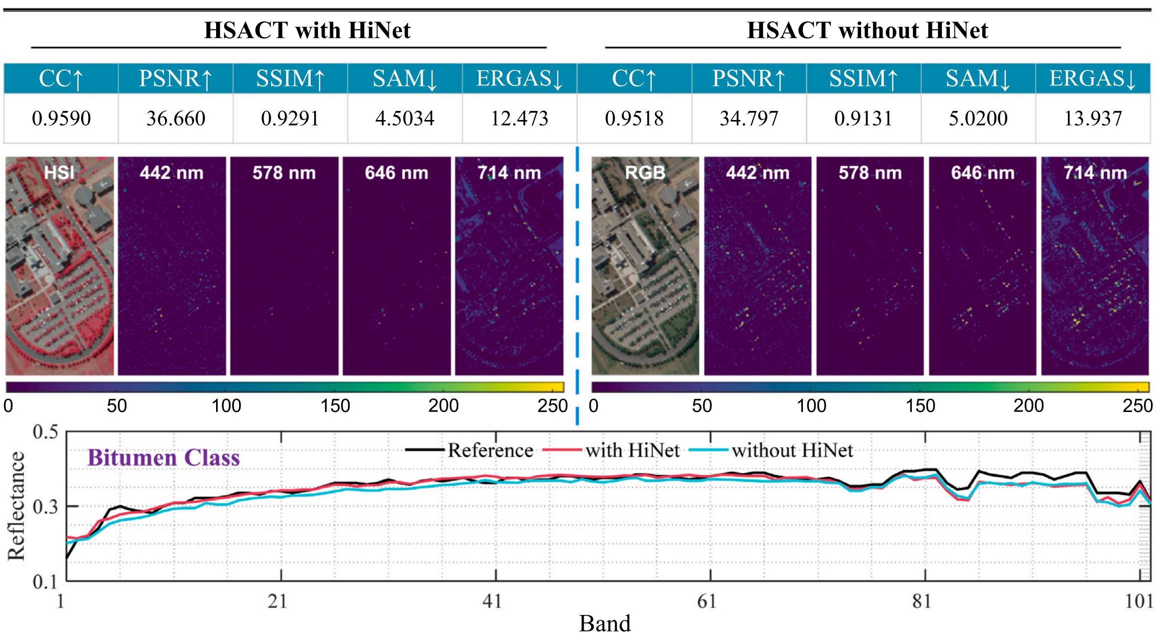

Fig. 7. SSR performance of the proposed HSACT method with HiNet and without HiNet in terms of quantitative indicators, spatial reconstruction error and spectral reconstruction error.

in the SSR task of the HSACT. The HiNet is able to perform lossless feature aggregation with multiple receptive fields. As it can be seen in Table 1, the SSR performance of the HSACT decreases when HiNet is removed. In addition, to further demonstrate the contribution of HiNet to the reconstruction performance of the proposed HSACT method on the University of Pavia scenario, the spatial and spectral reconstruction errors of HSACT with and without HiNet are presented in Fig. 7, it can be seen that the HSACT method with HiNet has higher spatial fidelity and spectral recovery ability. This indicates the advantage of hierarchical network architectures when conducting SSR of RGB images.

3.5.3. Spectral-spatial loss function

The effect of the SSL module in Section 2.4 is also demonstrated. The HSACT training process only relies on the constraints of the rate L_1 loss when the $L_{\text{spe-spa}}$ loss is ablated. The experimental results presented in Table 1 demonstrate that a performance degradation occurs in the SSR of the HSACT method. The reason is that the $L_{\text{spe-spa}}$ uses the spatial and spectral information of the reconstructed HSI as constraints to guide network learning, and it can effectively restore the spectral-spatial details of the image.

3.6. Comparisons

In this section, we compare the SSR performance of seven contrasting methods with the proposed HSACT method via quantitative experiments on the University of Pavia, Washington DC, and Wenxian datasets. As illustrated in subsection Section 2.1, for the fairness of the experimental comparison, 1024 RGB-HSI image pairs are randomly selected from each scene as the training set, and the rest are used as the testing set. In addition, it is worth mentioning that the traditional methods (i.e., Arad and J-SLoL) are repeated 10 times under the same experimental scheme and we report the mean results. The deep learning methods (i.e., HSRNet, MST++, UGCT, LRTN, HPRN, and the proposed HSACT) are trained for 200 epochs under the same training data batch to take the results of the last epoch. Details of spectral reconstruction, spatial fidelity, classification validation, and related discussions are presented below.

3.6.1. Evaluation of spectral reconstruction for SSR

Fig. 8 gives experimental results in terms of spectral reconstruction for all SSR methods on the University of Pavia, Wshington DC, and Wenxian datasets. Three regions of interest (ROIs) are selected for each of the scenes, i.e., bitumen, bare soil, and gravel ROIs for the University of Pavia dataset; buildings, water and asphalt road ROIs for the Washington DC dataset; and cropland, mudflat and built-up areas ROIs for the Wenxian dataset. Note that, taking into account the local radiance bias, we use the mean spectra of the ten pixels in each ROI in the reference HSI and the reconstructed HSI for visualization purposes. Looking at Fig. 8(a), we conclude that the dictionary learning approach (i.e., Arad) shows large reconstruction bias in the spectral

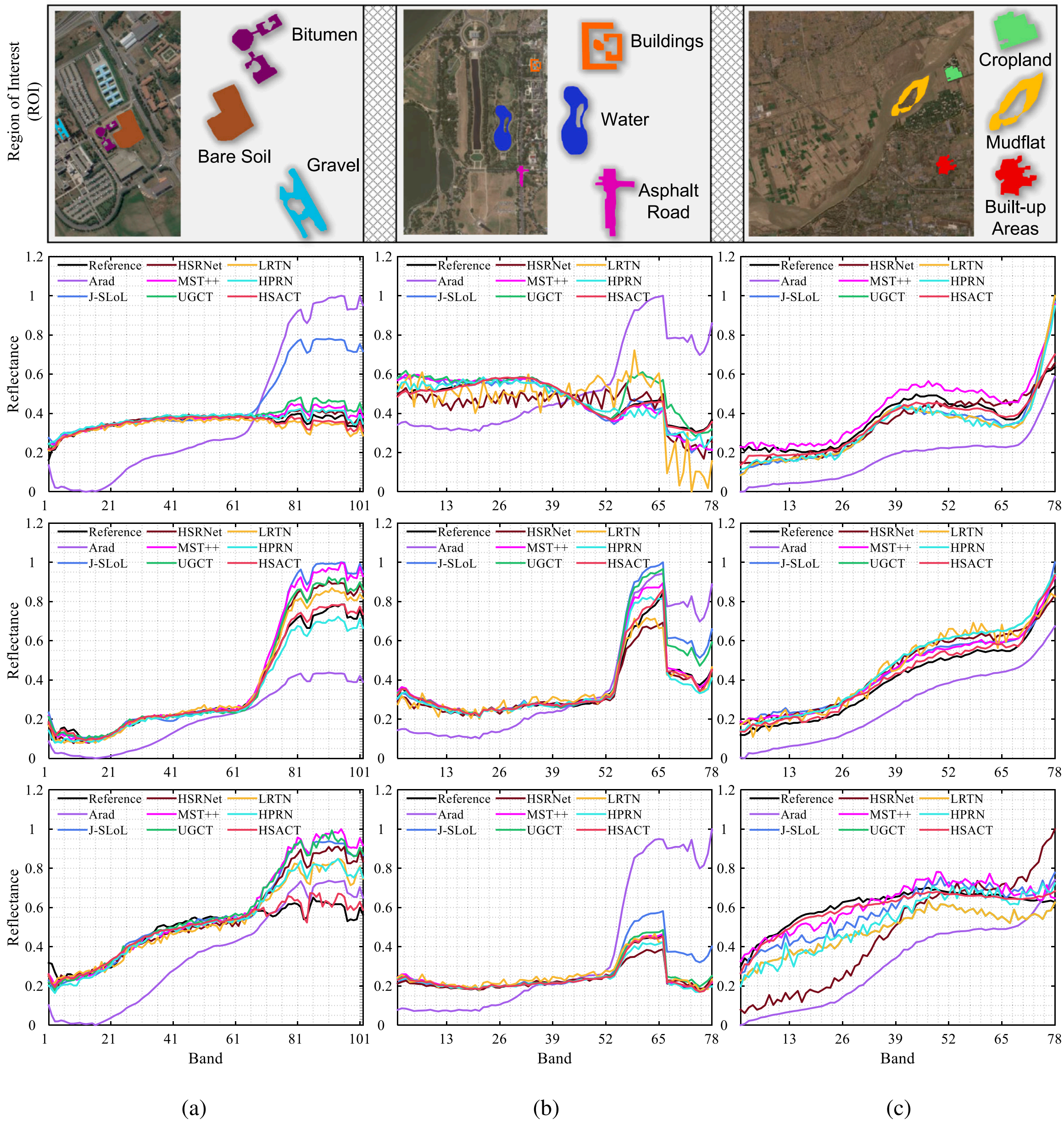

Fig. 8. Reflectance of various ground covers from different ROIs for each SSR method using the University of Pavia, Washington DC, and Wenxian datasets. (a) From top to bottom: bitumen, bare soil, and gravel features from the University of Pavia dataset. (b) From top to bottom: buildings, water, and asphalt road features from the Washington DC dataset. (c) From top to bottom: cropland, mudflat, and built-up area features from Wenxian dataset.

reconstruction of three ROIs from the University of Pavia. The J-SLOL exhibits better spectral restoration performance than Arad on three ROIs, but obvious deviations can be clearly observed in the bitumen ROI. In contrast, the SSR performance of the deep learning methods is better than that of the traditional methods, but each deep learning method achieves different degrees of spectral reconstruction bias in the 70–102 band interval. It can be seen from Fig. 8(a) that the proposed HSACT method (in red) is closer to the reference spectrum (in black) compared to other methods over the three ROIs.

Furthermore, similar phenomena can be observed in the Washington DC [see Fig. 8(b)] and Wenxian [see Fig. 8(c)] datasets. The traditional methods suffer from apparent spectral reconstruction errors over the entire wavelength range. The deep learning methods (i.e., HSRNet, MST++, UGCT, LRTN, and HRPN) outperform the traditional methods in terms of spectral reconstruction, but also present different biases (that is, below or above the reference spectrum). To sum up, the proposed HSACT method shows promising spectral reconstruction performance with respect to other deep learning approaches on the University of Pavia, Washington DC, and Wenxian datasets. This implies that the proposed HSACT method can effectively reduce the spectral distortion and improve the overall performance of SSR for remote sensing RGB images.

3.6.2. Evaluation of spatial fidelity for SSR

Table 2 presents the quantitative experimental results of seven comparison methods and the proposed HSACT in terms of spatial and spectral information recovery. As for the spatial fidelity of the spectral images, the CC scores of the traditional methods are lower than 0.8, and the highest value of the SSIM metric is 0.8209, which indicates that serious spatial degradation occurs in the reconstructed HSIs (see Fig. 9). Among the benchmark methods based on deep learning, the better performer is HSRNet with CC and SSIM metrics achieving 0.9437 and 0.9006, respectively, and the worse performer is LRTN with CC and SSIM of 0.9155 and 0.8598, respectively. In contrast, the CC and

Table 2
Spectral super-resolution results obtained for the University of Pavia scene by the Arad [30], J-SLoL [41], HSRNet [42], MST++ [43], UGCT [44], LRTN [45], HRPN [38], and the proposed HSACT method. The bold type is used for the optimal indicator, and the Italic underline is used for the sub-optimal indicator.

Metric		Various SSR methods on the University of Pavia scene							
		Arad	J-SLoL	HSRNet	MST++	UGCT	LRTN	HRPN	HSACT
CC	↑	0.7382	0.7416	0.9437	0.9386	0.9252	0.9155	*0.9466*	**0.9590**
PSNR	↑	22.263	34.236	32.613	34.488	*35.737*	30.751	35.237	**36.660**
SSIM	↑	0.7012	0.8209	0.9006	0.8897	0.8939	0.8598	*0.9135*	**0.9291**
SAM	↓	14.066	8.9032	5.6005	5.8092	5.8832	6.4174	*5.1257*	**4.5034**
ERGAS	↓	85.230	22.838	16.567	15.085	15.354	19.291	*14.116*	**12.473**

Table 3
Spectral super-resolution results obtained for the Washington DC scene by the Arad [30], J-SLoL [41], HSRNet [42], MST++ [43], UGCT [44], LRTN [45], HRPN [38], and the proposed HSACT method. The bold type is used for the optimal indicator, and the Italic underline is used for the sub-optimal indicator.

Metric		Various SSR methods on the Washington DC scene							
		Arad	J-SLoL	HSRNet	MST++	UGCT	LRTN	HRPN	HSACT
CC	↑	0.7278	0.9113	0.9134	0.9291	0.9232	0.9248	*0.9452*	**0.9537**
PSNR	↑	21.392	36.972	32.144	37.189	37.147	30.518	*35.471*	**37.421**
SSIM	↑	0.5156	0.8543	0.8776	*0.9267*	0.9102	0.8389	0.8992	**0.9297**
SAM	↓	20.208	12.681	11.087	*7.0152*	9.3961	10.371	7.6561	**6.2005**
ERGAS	↓	149.94	27.826	21.921	*19.211*	20.774	21.430	17.261	**15.844**

SSIM metrics of the proposed HSACT method achieved optimal results of 0.9590 and 0.9291, respectively. As for the spectral accuracy of the spectral images, the SAM scores of the Arad and J-SLOL reached 14.066 and 8.9032, respectively, while the SAM scores of the competing methods based on deep learning are all below 8.0 [e.g., UGCT (SAM = 5.8832) and HRPN (SAM = 5.1257)]; the traditional SSR methods exhibit weak spectral reconstruction capabilities relative to the deep learning benchmark methods. The proposed HSACT achieved 4.5034 in terms of the SAM metric, which is 49.42% higher than the best traditional method and 23.45% higher than the best deep learning method, showing excellent spectral information recovery in the SSR task. Regarding the spectral-spatial recovery metrics, the proposed HSACT method is still able to maintain the highest performance compared to traditional and deep learning-based benchmark methods.

Table 3 provides the experimental reports of all SSR methods on Washington DC data, where all SSR methods are further validated in terms of SSR performance using the dataset with a spectral range of 400–1000 nm containing 78 spectral bands. It can be seen that the J-SLOL method achieves the best SSR performance among traditional methods, but its reconstruction ability is still weaker than that achieved by the deep learning methods. Specifically, the five deep learning comparison methods performed relatively well; for example, the SeqCnv-typed HSRNet method achieved SAM = 11.087 and CC = 0.9134; the MST++ method (that belongs to EnDeCoder) obtained SAM = 7.0141 and CC = 0.9287; and the Transformer-based UGCT method obtained SAM = 9.396 and CC = 0.9232. The HSACT method (using a hierarchical structure) achieved SAM = 6.2005 and CC = 0.9537, respectively, which fully demonstrates the advantages of the hierarchical network architecture over the SeqConv, EnDeCoder, and pure Transformer network architectures in the SSR task. It is worth mentioning that it can be observed from Table 3 that the proposed HSACT method has obvious advantages over the two deep networks based on prior learning (i.e., LRTN and HPRN). Furthermore, the spectral images for each method at different wavelengths are presented in Fig. 10, in which it can be observed that the proposed HSACT method exhibits superior SSR performance in terms of spatial fidelity and spectral restoration compared to the traditional and advanced comparative methods.

In addition to the above two validation experiments on simulated datasets, we tested the performance of all SSR methods on real scenarios (i.e., the Wenxian dataset). Importantly, the RGB images in the Wenxian dataset are obtained from Sentinel-2 data and the HSI is obtained from GF-5 data (see Section 2.1), and the images are captured at different times, thus it is inevitable that there are slight variations due to weather changes and different sensor acquisitions. Table 4 presents the objective evaluation metrics for all the SSR methods. It can be seen that most learning-based benchmark methods perform better than the traditional methods, but the partial reconstruction metric of LRTN is slightly lower than that of the J-SLoL method, which means that the application of LRTN method in real scenarios has low robustness. In contrast, the proposed HSACT outperforms all the competitive methods in terms of spatial, spectral, and spectral-spatial composite metrics. Similarly, a visualization of all the SSR methods is given in Fig. 11, where it can be seen that HSACT still exhibits better spatial detail reconstruction ability than the other tested methods. Overall, the HSACT approach deeply exploits the spectral and spatial reconstruction capabilities of the network for SSR tasks by embedding superpixel semantic features in a learnable way while employing a novel hierarchical architectural design. Moreover, our HSACT achieves excellent performance compared to all the competitive methods in both experiments on simulated and real datasets, which proves the effectiveness and practicality of HSACT in the SSR task.

3.6.3. *Results on classification validation*

Here we utilize the classification results of the support vector machine (SVM) [46] classifier on different sources for the same scene to further validate the quality of the HSACT-reconstructed HSIs. Note that the classification results of the SVM are averaged over 10 times, and the training set in each scene is composed of 180, 60, and 120 labeled pixels that are randomly selected from each scene. The overall accuracy (OA), average accuracy (AA) and Kappa coefficients are used as the evaluation metrics of classification performance. The reconstructed HSI obtained from SSR by the competition methods and the proposed HSACT method are denoted as method abbreviation.

As shown in Table 5, the SVM is able to establish more effective classification criteria on the reconstructed-HSIs compared to RGB images for ground covers interpretation. For instance, the SVM applied to the reconstructed-HSI by the HSACT method respectively improves the OA, AA, and Kappa by 10.56, 5.43, and 11.8 over the SVM applied to the RGB of the University of Pavia dataset. Furthermore, as can be seen from Table 5, the HSACT method obtained more reliable reconstruction results based on classification accuracy in terms of similar SSR methods. Similarly, it can be found that the classification performance of the SVM has been effectively improved on Washington DC and Wenxian scenes by using HSI reconstructed by the HSACT method. In addition, Fig. 12 gives a visualization of the interpretation maps obtained by

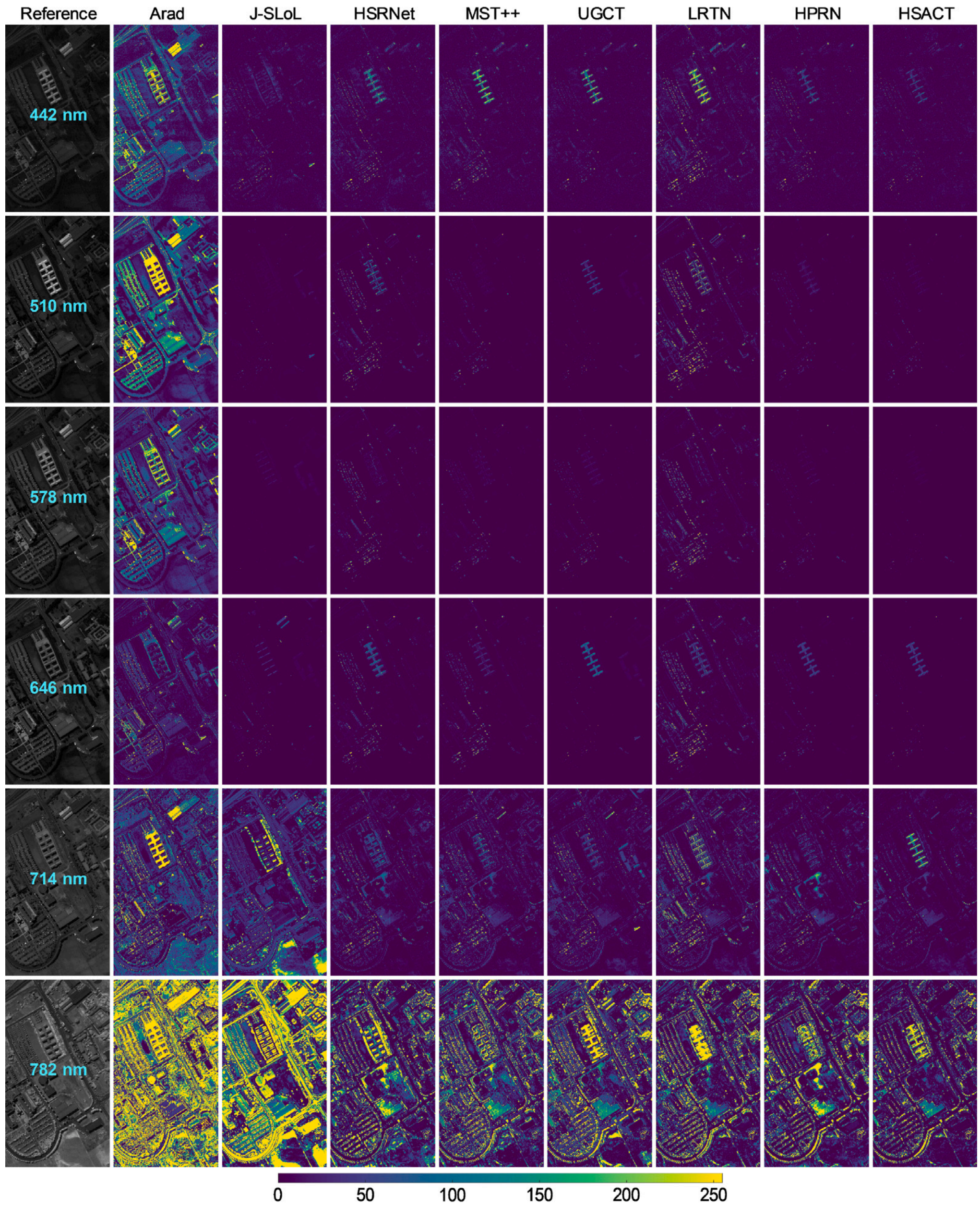

Fig. 9. Square differences for the University of Pavia dataset. Along 442, 510, 578, 646, 714, and 782 nm, the square differences between the image and the ground truth (GT) are given. The first column is the GT (CC), and the second to last columns are the results of Arad (0.7885), J-SLOL (0.7986), HSRNet (0.9502), MST++ (0.9441), UGCT (0.9357), LRTN (0.9121), HPRN (0.9431), and the proposed HSACT method (0.9644), respectively.

Table 4
Spectral super-resolution results obtained for the Wenxian scene by the Arad [30], J-SLoL [41], HSRNet [42], MST++ [43], UGCT [44], LRTN [45], HRPN [38], and the proposed HSACT method. The bold type is used for the optimal indicator, and the Italic underline is used for the sub-optimal indicator.

Metric		Various SSR methods on the Wenxian scene							
		Arad	J-SLoL	HSRNet	MST++	UGCT	LRTN	HRPN	HSACT
CC	↑	0.7505	0.8336	0.8255	0.8569	<u>*0.8614*</u>	0.7852	0.8577	**0.8653**
PSNR	↑	24.378	32.122	32.892	33.263	<u>*33.769*</u>	32.139	33.729	**33.911**
SSIM	↑	0.5838	0.8378	0.8526	0.8648	<u>*0.8685*</u>	0.8304	0.8637	**0.8698**
SAM	↓	15.203	4.8436	5.2564	4.1751	<u>*3.7506*</u>	5.5832	4.1478	**3.8283**
ERGAS	↓	228.88	21.912	20.515	19.869	<u>*18.026*</u>	22.247	18.305	**17.753**

the SVM for the reconstructed-HSIs (obtained by traditional and deep learning-based) and RGB images, respectively. It can be observed that the classification maps obtained by the SVM for the reconstructed-HSI by the HSACT method are closer to the ground truth [see the first column in Fig. 12]. Therefore, this experiment further proves the effectiveness of the HSACT method in the SSR task, and also confirmed that the HSIs reconstructed by the HSACT method can dramatically improve the feature interpretation capability of the classifier. It is worth

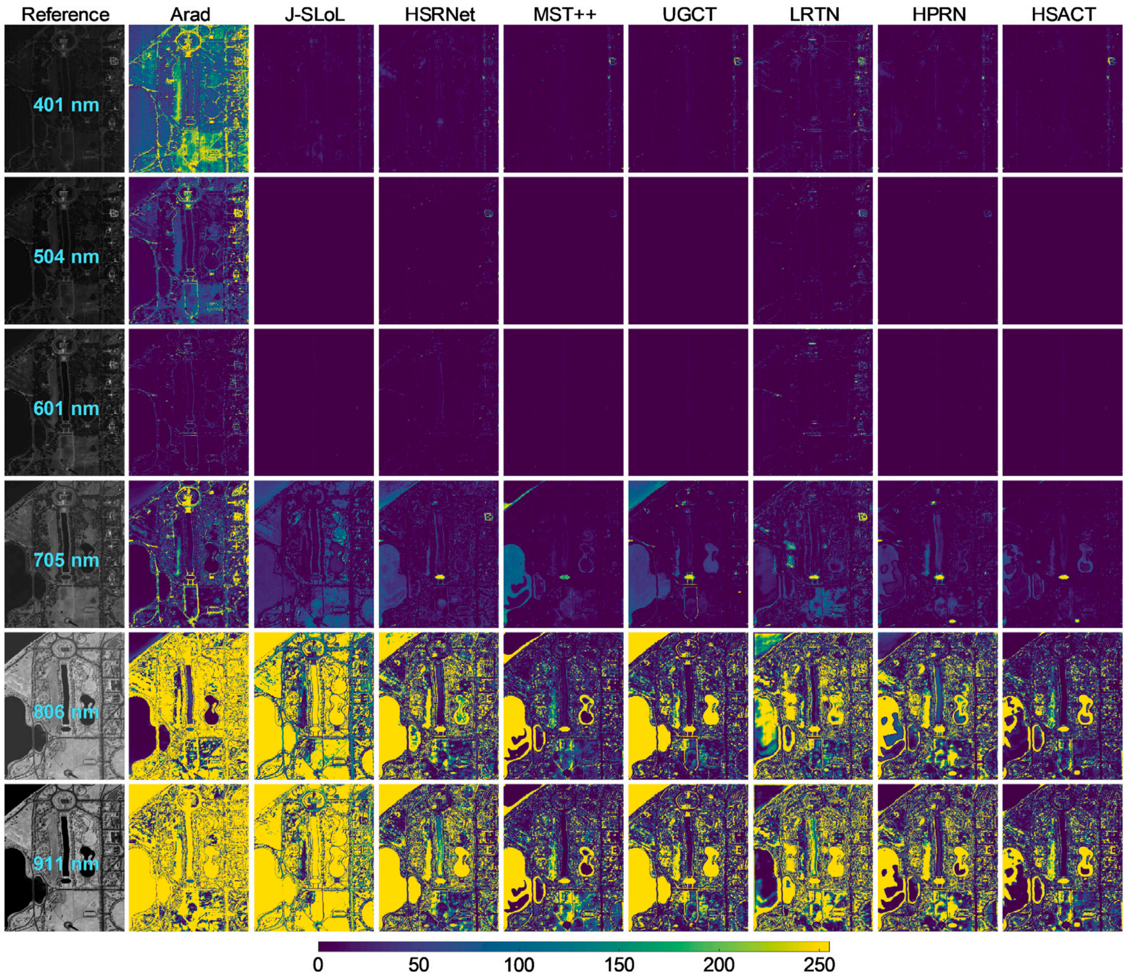

Fig. 10. Square differences for the Washington DC dataset. Along 401, 504, 601, 705, 806, and 911 nm, the square differences between the image and the ground truth (GT) are given. The first column is the GT (CC), and the second to last columns are the results of Arad (0.7335), J-SLOL (0.8797), HSRNet (0.8894), MST++ (0.9078), UGCT (0.9001), LRTN (0.9205), HPRN (0.9411), and the proposed HSACT (0.9502) method, respectively.

Table 5
Classification accuracies (in percentage) obtained by the SVM classifier using the RGB image and the HSI reconstructed by the competition methods and the proposed HSACT method for various scenes. The scores in the parentheses are standard deviations.

Metric		Different data sources for different scenarios								
		RGB	Arad	J-SLoL	HSRNet	MST++	UGCT	LRTN	HPRN	HSACT
University of Pavia	OA (%)	62.69	59.35	60.52	69.27	66.13	66.52	69.12	<u>*71.53*</u>	**73.25**
	AA (%)	66.78	61.77	65.02	71.35	69.00	68.60	70.82	<u>*71.41*</u>	**72.71**
	Kappa (%)	53.91	50.07	51.67	61.69	58.34	58.46	61.06	<u>*64.06*</u>	**65.71**
Washington DC	OA (%)	76.00	76.10	78.37	78.42	81.14	76.37	<u>*83.01*</u>	80.54	**83.70**
	AA (%)	61.82	62.02	63.52	66.42	67.98	61.98	<u>*69.04*</u>	65.46	**69.67**
	Kappa (%)	65.25	65.79	68.71	69.78	73.28	66.14	<u>*75.60*</u>	71.61	**76.76**
Wenxian	OA (%)	70.86	72.30	73.18	72.96	76.56	<u>*77.81*</u>	76.48	76.45	**78.54**
	AA (%)	66.34	66.08	69.24	68.89	72.53	<u>*73.66*</u>	71.23	72.98	**73.04**
	Kappa (%)	63.94	65.39	66.67	66.00	70.84	<u>*72.38*</u>	70.81	71.00	**73.26**

mentioning that the HSACT-reconstructed HSIs are not only beneficial to the classification task of remote sensing observations, but are also suitable for other tasks (e.g., anomaly detection and change detection).

3.7. Discussion

The reliability of the proposed HSACT method is discussed in detail from different viewpoints in this section, including the balance parameter in our designed spectral-spatial loss function, the model comprehensive performance, and the model complexity.

3.7.1. Balance parameter

Table 6 investigates the effect of the balance parameter α in the spectral-spatial loss function on the performance of the proposed HSACT method using the University of Pavia dataset. The values of α are preset to $\{1, 0.1, 0.001, 0\}$. $\alpha = 1$ means that the spectral and spatial loss values of the reconstructed HSI are as important as the $\mathcal{L}_1$ loss values during model training, while $\alpha = 0$ implies that there is only $\mathcal{L}_1$ loss. It can be observed that the CC, SAM, and ERGAS of HSACT have a tendency to increase and then decrease as α changes according to the preset interval. This demonstrates that constraining the network using the spectral-spatial information in the HSI is beneficial to the performance of the HSACT method.

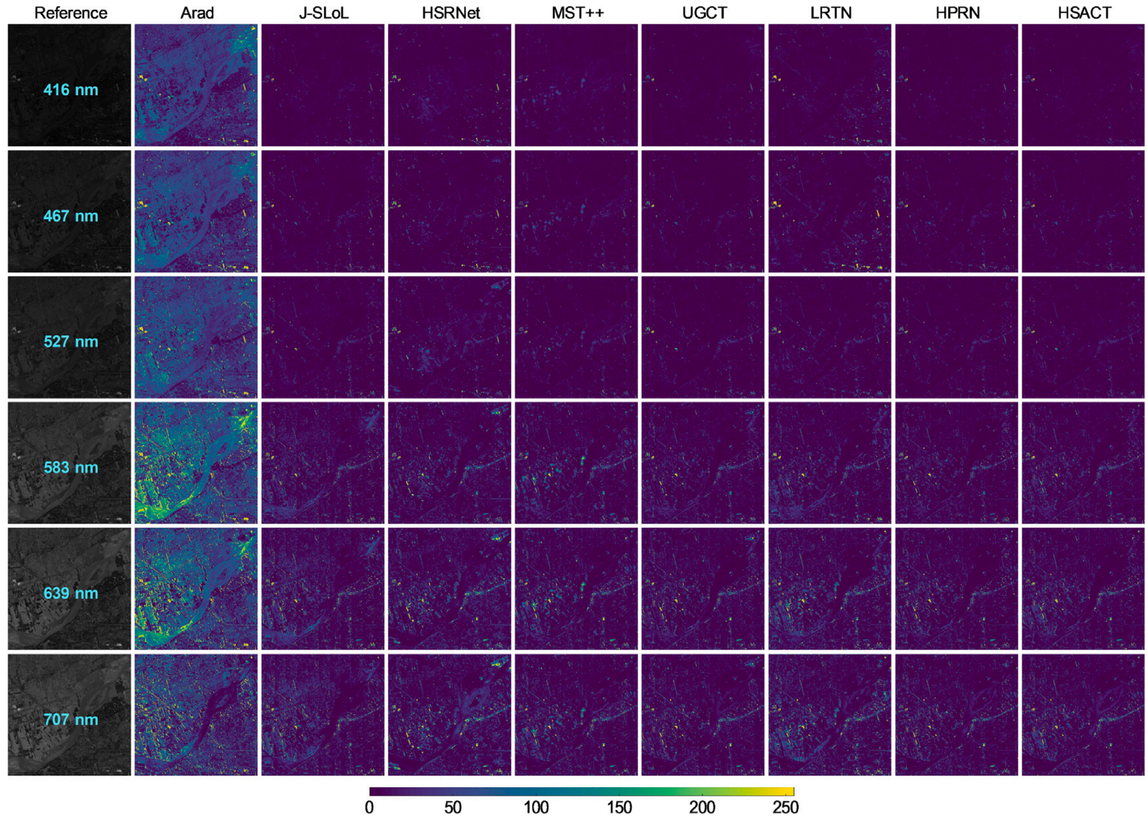

Fig. 11. Square differences for the Wenxian dataset. Along 416, 467, 527, 583, 639, and 707 nm, the square differences between the image and the ground truth (GT) are given. The first column is the GT (CC), and the second to last columns are the results of Arad (0.7814), J-SLOL (0.8413), HSRNet (0.8377), MST++ (0.8685), UGCT (0.8734), LRTN (0.7956), HPRN (0.8631), and the proposed HSACT (0.8789) method, respectively.

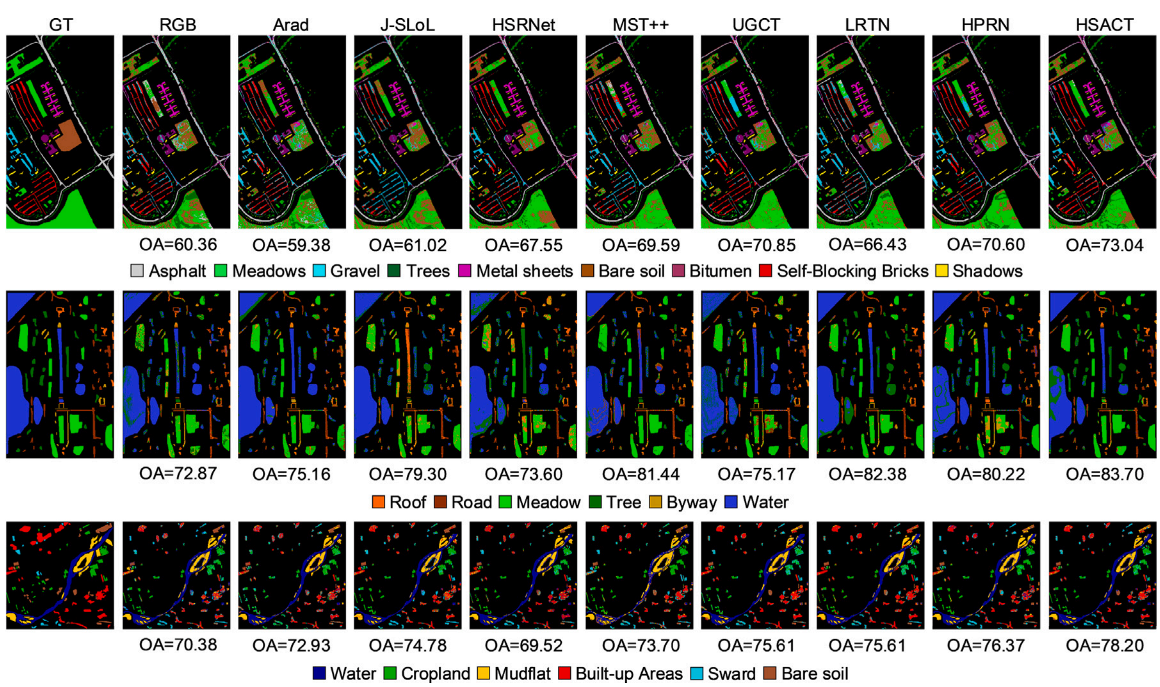

Fig. 12. Classification results obtained by SVM using RGB images and the HSIs reconstructed by the competition methods and proposed HSACT method on various scenarios. From the first row to the last row is University of Pavia, Washington DC, and Wenxian scenario.

3.7.2. Comprehensive performance

As shown in Fig. 13, to evaluate the comprehensive performance (including accuracy and number of parameters) of deep learning-based SSR methods, we analyze the CC, SAM and number of parameters of these methods on the University of Pavia dataset. The closer that the points are to the upper left corner indicates the better SSR performance, and the smaller the point size means a more lightweight model. It can be seen that the HPRN method performs better than other methods

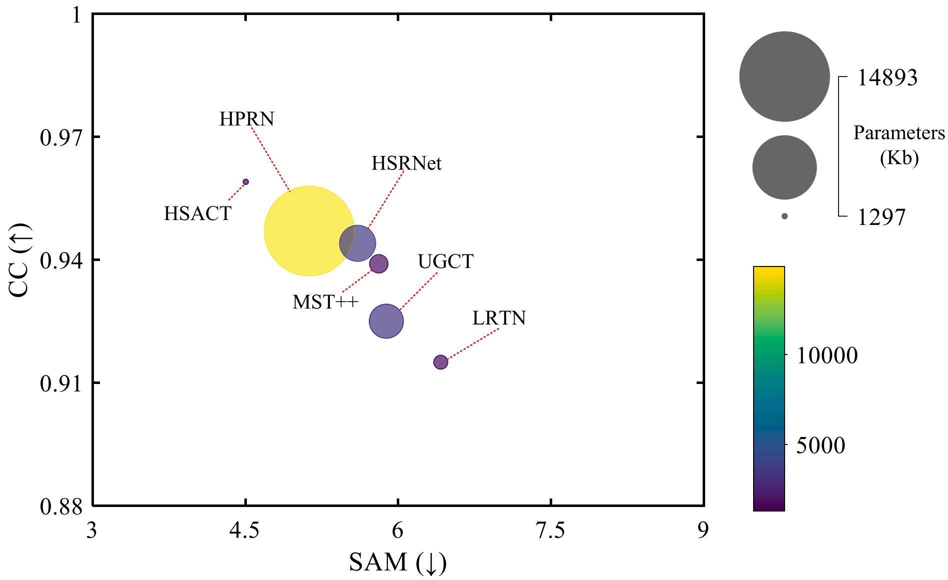

Fig. 13. Performance analysis of deep learning methods when applied to the University of Pavia dataset.

Table 6
SSR performance of the proposed HSACT method with various values of the balance parameter α on University of Pavia scene.

α	Washington DC			
	1	1×10^{-1}	1×10^{-3}	0
CC ↑	0.9489	0.9532	0.9677	0.9540
SAM ↓	6.3554	6.6560	5.0454	6.2771
ERGAS ↓	16.415	15.909	13.344	15.026

Table 7
Execution performance of the compared methods in the SSR task.

Method	Training (s)	Testing (s)	Param. (MB)	FLOPs (GB)
Arad	16.431	2.106	N/A	N/A
J-SLoL	225.65	1.330	N/A	N/A
HSRNet	1141.7	1.338	3.453	50.407
MST++	868.30	1.305	1.825	9.1991
UGCT	7968.9	1.413	3.229	52.907
LRTN	2128.8	8.820	1.580	4.9205
HPRN	4888.1	6.708	14.89	243.22
HSACT	736.11	1.317	1.297	21.823

but, in contrast, the HSACT method outperforms the HPRN method in terms of both SSR performance and the number of model parameters. This indicates that HSACT is a lightweight model with good SSR performance.

3.7.3. Model complexity

Table 7 reports the performance efficiency achieved by all SSR methods considered in this work with the University of Pavia dataset, including training time, testing time, number of parameters, and floating-point operations per second (FLOPs). According to the table, the HSACT method achieves the better performance in terms of training time, testing time, and number of parameters than most deep-learning-based methods. This fully demonstrates the performance efficiency of the proposed HSACT method in SSR from RGB image to HSIs.

4. Conclusions and future lines

In this paper, we propose a new hierarchical semantic-aware CNN-Transformer (HSACT) to reconstruct HSIs based on the corresponding remote sensing RGB images. The backbone network of the proposed method is designed as a hierarchical architecture (instead of the traditional encoder–decoder and sequential convolution architectures) to associate wavelength constraints for spectral image estimation, and uses a semantic-aware CNN-Transformer. The main advantages of our method can be summarized as follows: (1) it adopts the idea of hierarchical networks, which include spectral-spatial feature aggregation and network multiplexing with weight sharing; (2) superpixel semantics no longer stands alone as a data pre-processing module, but as a learnable semantic embedding which can be included in plug-and-play fashion. Our experimental results, performed on simulated and real datasets, demonstrated that the proposed HSACT method restores HSIs from remote sensing RGB images more effectively than other traditional and advanced SSR methods. This paper also released a new RGB-HSI dataset for SSR purposes (Wenxian), which allows us to conduct the first attempt to recover GF-5 HSI data from Sentinel-2 RGB data. However, there are two major limitations in the transfer of the proposed HSACT method to new scenarios. On the one hand, the preset parameters of superpixel blocks and the last layer of the network need to be fine-tuned in new scene applications due to the differences in image spatial structure and hyperspectral imaging. On the other hand, the HSACT method requires adaptive adjustments to the input and ISE in SSR tasks beyond the visible light spectrum.

In future work, we will focus on the deep learning paradigm of coupling expert knowledge with deep networks in SSR tasks, which mainly includes two directions: (1) embedding prior knowledge into the diffusion model [47], combining timestamp and prior information, accelerating the efficiency of network training, and improving inference time-consumption of diffusion models; (2) using deep networks to decouple traditional optimization models, promoting the reconstruction accuracy of traditional optimization models for SSR, and improving the interpretability of the network. In addition, we will further study the performance of the SSR model in environments with imaging equipment interferes and beyond the visible light spectrum.

CRediT authorship contribution statement

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data are available online from https://github.com/chengle-zhou/RGB-HSI_Wenxian. The codes are available online from https://github.com/chengle-zhou/HSACT.

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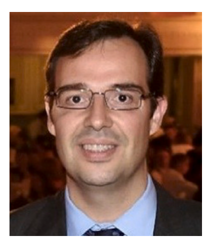

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